

High-Salt Binding Buffer Reduces DnaK Co-Elution During Nickel Affinity Purification of a Recombinant Collagen-Like Protein

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ABSTRACT

A 95-kDa His10-tagged collagen-like protein, ColX3, was purified from *Escherichia coli* using nickel-nitrilotriacetic acid resin. We tested whether increasing sodium chloride from 300 to 500 mM during lysis and binding would reduce co-elution of the 70-kDa chaperone DnaK. The higher-salt condition lowered the DnaK-to-ColX3 band-density ratio by 2.7-fold while causing a modest decrease in recovered target protein from 5.6 to 5.1 mg/L culture. ATP-Mg²⁺ treatment before loading did not further improve purity. These data support high-salt binding as a simple process adjustment when DnaK is carried through nickel affinity purification by a client protein.

Keywords: Ni-NTA, DnaK, recombinant protein, sodium chloride, purity

INTRODUCTION

Recombinant collagen-like proteins can expose extended hydrophobic or non-native regions during bacterial expression. Cytosolic chaperones may bind these regions and follow the target through affinity purification even when the chaperone itself lacks an affinity tag. DnaK is a common 70-kDa co-purifying protein in *E. coli* expression systems.

This study asked whether ionic strength could reduce DnaK co-elution without eliminating recovery of ColX3. The primary comparison was 300 versus 500 mM sodium chloride during lysis, binding, and wash. ATP-Mg²⁺ treatment was included as a secondary condition because nucleotide-dependent release has been reported for some DnaK-client complexes. The study was designed as a process comparison rather than a mechanistic analysis of DnaK phosphorylation or expression.

MATERIALS AND METHODS

Expression and lysis

His10-ColX3 was expressed in *E. coli* BX21 from a synthetic plasmid. Cultures were induced at an optical density of 0.8 with 0.10 mM IPTG and incubated for 18 h at 16 C. Cell pellets from three independent 1-L cultures were divided equally between the two salt conditions.

Pellets were resuspended in 50 mM Tris-HCl, pH 8.0, 10 mM imidazole, and either 300 or 500 mM NaCl. Lysates were produced by two passes through a high-pressure homogenizer and clarified at 20,000 x g for 30 min.

Affinity purification

Clarified lysate was loaded onto 5 mL of Ni-NTA resin at 0.5 mL/min. Resin was washed with 12 column volumes containing 60 mM imidazole and eluted with 500 mM imidazole. In the ATP condition, clarified lysate was incubated with 5 mM ATP and 6 mM MgCl2 for 20 min on ice before loading.

Target recovery was measured by bicinchoninic acid assay after densitometric correction for purity. DnaK identity was confirmed with a synthetic anti-DnaK reference reagent. Band-density ratios were calculated from reducing SDS-PAGE.

Statistics

Each value represents three independent cultures. Two-sided Welch tests were used for pairwise comparisons. A difference was considered descriptive when the nominal P value was below 0.05; no correction was required because the two primary outcomes were prespecified.

Condition	NaCl	ATP/Mg2+	Wash imidazole
Low salt	300 mM	No	60 mM
High salt	500 mM	No	60 mM
High salt + ATP	500 mM	5/6 mM	60 mM

Table 1. Prespecified purification conditions.

RESULTS

Target recovery

ColX3 remained predominantly soluble in both salt conditions. The mean recovered target was 5.6 +/- 0.4 mg/L at 300 mM NaCl and 5.1 +/- 0.3 mg/L at 500 mM NaCl. The 9% reduction at higher salt was reproducible but small relative to the improvement in purity. Flow-through fractions did not show a large increase in unbound ColX3.

DnaK co-elution

The DnaK-to-ColX3 band-density ratio fell from 0.62 +/- 0.08 at 300 mM NaCl to 0.23 +/- 0.05 at 500 mM NaCl, a 2.7-fold reduction. DnaK was still detectable, so the high-salt condition did not completely disrupt the complex. Adding ATP-Mg2+ before loading produced a ratio of 0.21 +/- 0.06, which was not detectably different from high salt alone.

Other impurities

Bands near 45 and 60 kDa were not consistently affected by salt concentration. Endotoxin and enzymatic activity were not measured because ColX3 is a structural protein and the experiment was limited to capture purity.

Condition	ColX3 yield (mg/L)	DnaK/ColX3 ratio
300 mM NaCl	5.6 +/- 0.4	0.62 +/- 0.08
500 mM NaCl	5.1 +/- 0.3	0.23 +/- 0.05
500 mM + ATP/Mg2+	5.0 +/- 0.5	0.21 +/- 0.06

Table 2. Recovery and co-elution outcomes. Values are mean +/- SD, n=3.

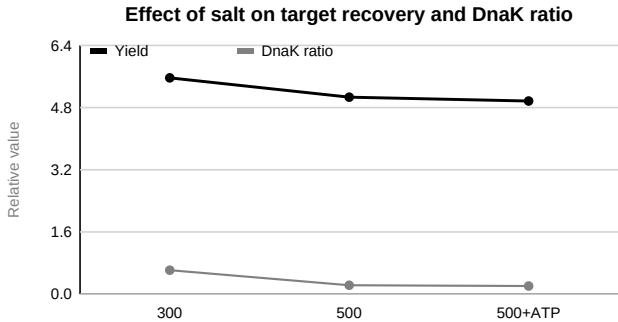


Figure 1. High salt reduced the relative DnaK signal with only a modest loss of target recovery.

DISCUSSION

Increasing NaCl to 500 mM improved the apparent purity of His10-ColX3 by selectively reducing DnaK co-elution. Because ColX3 binding to the resin was largely retained, the effect is more consistent with weakening of a noncovalent DnaK-client association than with general loss of nickel affinity.

ATP-Mg²⁺ did not add a measurable benefit under the tested conditions. This negative result does not establish that the DnaK complex is nucleotide-insensitive; the incubation time, nucleotide state, and presence of co-chaperones were not varied. The result only shows that the specified pretreatment was not useful as an added process step.

LIMITATIONS

The work used one recombinant protein, one expression host, and a single nickel resin. DnaK phosphorylation, client stoichiometry, and binding kinetics were not measured. Band densitometry provides relative rather than absolute impurity concentration.

CONCLUSION

A 500 mM NaCl lysis and binding buffer reduced DnaK co-elution by 2.7-fold while retaining approximately 91% of ColX3 recovery. ATP-Mg²⁺ pretreatment did not improve the high-salt process.

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The following references are fictional and are included only to test reference-section handling.

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